



**PENTECOST
UNIVERSITY**

TRANSFORMATION AND SERVICE

ASSIGNMENT

NAME:

NII ARMATEY MENSAH JOEL

STUDENT ID:

PUIT/24110004

PROGRAM:

BSC. INFORMATION TECHNOLOGY

COURSE:

DATA COMMUNICATION

LEVEL: 200 (2ND SEMESTER)



EVOLUTION OF SATELLITE COMMUNICATION AND ITS ROLE IN THE MODERN GLOBAL DATA INFRASTRUCTURE.

Abstract

In this age of machines, Satellite data transmission has been the focus of engineering and the world's dream. Data is being transmitted from earth to space. In this essay, we talk about the design, architecture and the intricate processes of making the satellite systems one of the most useful of this information revolution era. The evolution of satellites from its origins and development from relay stations into the load bearing structures for the global cyberspace is highlighted. This then lead to discussion of the physics of orbit mechanics in terms of Low Earth Orbit, Medium Earth Orbit and Geostationary Orbit First. The Geostationary Orbit is a discussion of the frequency spectrum management which determine the system capacity. The theme of this article is a technical discussion of High Throughput Satellite and the spot beam technology (aka satellite zooming) which has radically changed the economics of broadband service from space. We're awash in invisible waves; it gets more crowded every day. A way to cope with more data? Beaming smaller spots like a spotlight (instead of the floodlight). These "spots" boost costs initially but limitless opportunities later on. These satellites, unlike cables laid to ground or hung from poles, ignore geography - they fly over oceans and mountains. But delays in getting replies lag behind fiber and too many users jam the system. Even with clear skies, speeds are still slower. Eventually satellites will interweave with the new 6G Earthly networks - a trend fortified via intelligent automation led by artificial intelligence. Far from just complementing terrestrial networks, they now play a significant role in how global networks work. Sources are reliable journals, engineering publications (e.g. IEEE), and industry reports. The points are laid out gradually, and demonstrate the increasing role of space technologies in common networks.

Table of Contents

1. Introduction	5
2. How Satellites Communicate in Space.....	7
2.1 Near Earth, Mid Earth and Geostationary Orbits	7
2.2 Frequency Spectrum Management	9
2.3 Modifying Signals and Strengthening Connections	11
3. Satellite Gets Closer and Closer Accurate Drop.....	13
3.2 Frequency Reuse Architectures	14
3.3 Focused Coverage using Spot Beams.....	15
4. Satellite vs. Terrestrial Data Communication: Which is Best?	16
5. Problems Sending Data From Satellites	18
5.1 Atmospheric Attenuation	18
5.2 Propagation Latency	19
5.3 Space junk and space safety	19
5.4 Cybersecurity in Space	20
6. Future Trends.....	21
6.1 Hooking up to 6G	21
6.2 AI-powered smart satellites	23
7. Conclusion.....	25
References	27

1. Introduction

Satellites are the way the world communicates, probably more than anything else - even if we don't realize it. They are up in the sky, but we can't see them, and we rely on them for our everyday life. They are not ground-based, but up in the sky - connecting voices from across the world. They bring live broadcasts from around the world, bringing events to you. Timestamps on financial transactions are perfectly synchronized as signals bounce off them. They make maps know you to within a few metres with their ticking. No old-school earth bound and cabled system ever achieved this accuracy. Their peaceful operation obscures their creation: ingenious people laboring away in secret for years. Thoughts about orbiting objects began as early as before the computer was invented. While some envisioned shiny balls orbiting the Earth, others still had slide rules.

In 1945, Arthur C. Clarke published a paper in *Wireless World* proposing three relay stations in the sky over the equator could provide global signal connections - always in the same spot in the sky (Clarke, 1945). That notion? Not only was it fresh, it shocked the experts who hadn't considered an object being above a moving spot on Earth, by moving to keep pace with the Earth. But it kept nagging people - gradually warming over the decades. In the late 1960s, the theory became reality as satellites were launched. Enter Sputnik 1, flung into orbit by the Russians in October 1957 - that demonstrated it was possible. In 1960, NASA launched Echo 1 which reflected radio waves back to Earth. Though tiny, it was a start. Two years later Telstar 1 was launched by AT&T and Bell Labs. It passed the signal on, rather than reflected it. It was a change that was more significant than we realized. It was the key to TV being transmitted across the ocean live - images were sent across space to our screens. We saw it ongoing, across the sea. Geographical distances were no longer so vast. In that moment something was revealed: satellites were no longer just for measuring things or weather. They started bringing sound and images, ideas - home. Not only measuring upwards, but also speaking down. (Whalen, 2002).

Launching Intelsat in 1964 cleared the way for countries to manage satellite communications in the future. Intelsat I (aka Early Bird) moved into orbit in 1965, and carried just 240 phone calls; tiny today but huge at the time (Pelton, 2012). But, over time, every generation of these high-altitude contraptions had better signals, antennas and channels. Ditches started playing with the signals in the 1980s to 1990s, which slyly enabled data to flow together with voice. By the turn

of this century, those same satellites brought the internet to places too unwieldy or distant for cables to go.

A decade ago, the bulky and expensive satellites far out in space were kings - today the small ones close in are rapidly becoming the norm. Rather than the big birds up in the sky, new constellations such as Starlink, OneWeb (Amazon is also planning on it's own) fill the sky with thousands of tiny satellites. These swarms seek to bring speedy internet almost like that found using cables, to places previously inaccessible. This is unprecedented change in this sector, issues of regulation, technological limits, space junk and light pollution are still all up for grabs.

Meanwhile, in space, while constellations change, another change took place. Rather than huge blankets of radio beamed down, satellites now shoot at the earth with tiny beams - hundreds of them. The beams all squeeze into the same frequency band, and don't get in each other's way. One satellite does all the data once thought impossible, reaching speeds of up to gigantic. It's called zoom-like because of the laser-like focus, not speed. This focus provides internet in-flight, ships at sea. Where once there were calls with static, there's now video in full colour. These links are low in power, high in accuracy. One satellite is enough to do what many satellites did before.

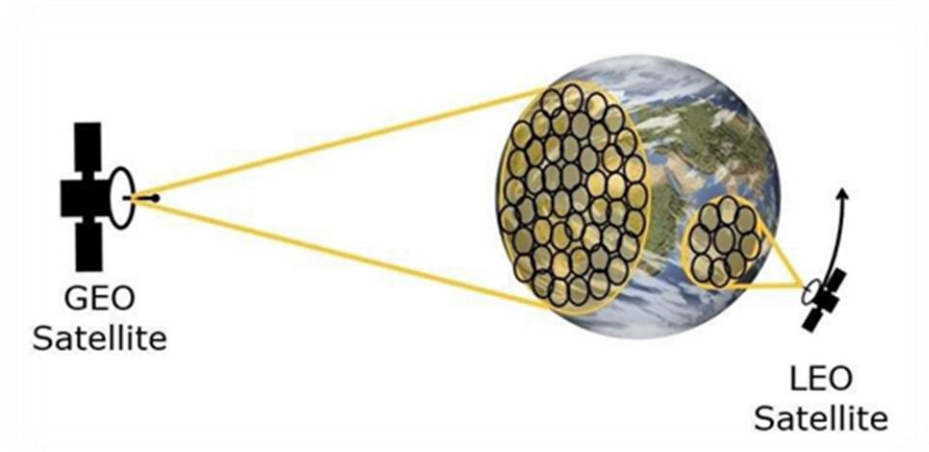
To understand how they work, you have to get into multiple disciplines. First off: orbits - how things move about, and decisions about where satellites are in space. There's radio waves - how signals behave in air and space, and blending in rules for using different frequencies. And then the signal design: techniques to squeeze information and combat interference, getting close to physical limits. Add to the mix design challenges - how to balance signal strength across long distances, ensure different networks don't interfere, route traffic around the globe. Each piece connects. The lines below break each piece down, and put them together. What emerges? It's a good picture of satellite communications today - and where they are heading.

Here's the table of contents. Let's start with Section 2, which sets out fundamentals - orbits of objects, allocation of radio frequency bands, as well as methods of encoding data. Then, Section 3 talks about high-throughput satellites - focusing on spot beams. Then, Section 4 compares satellite networks with terrestrial networks, side-by-side. Finally, Section 5 lifts the lid on tricky issues engineers grapple with, every day. Looking ahead, Section 6 discusses how satellites

could potentially connect with next-gen wireless networks, and also provides smart computing to space-based activities. Finally, Section 7 looks at the big picture - how satellite links can shape global information.

2. How Satellites Communicate in Space

2.1 Near Earth, Mid Earth and Geostationary Orbits



There are good reasons why most satellites have fixed orbits. How high up they are determining how much they cover, plus signal latency. It also determines lap time, types of space weather they encounter and requirements for fuel to stay in orbit. For global communications, it's typically one of three heights: near space laps, mid distance laps and geostationary spots. Each has their own trade-off. Over the years some applications found their place in one of these layers.

Way up there, almost 35,786 kilometers above the equator is Geostationary Orbit. Satellites there orbit the Earth in 23 hours, 56 minutes and 4 seconds - the same time it takes Earth to make a full rotation. Their orbit and Earth's rotation periods match so well they appear to be fixed in Earth's sky. This is important to communicate over long distances. Back on Earth, fixed pointed dishes are used on the receiving equipment, dishes that do not sweep and jiggle. This saves on costs and complications. A single satellite has wide beams covering nearly a third of the earth constantly. Distantly located villages are connected by the single vehicle in space.

On the way up, signals travelling to a GEO satellite cover a lot of distance. Traveling at the speed of light, radio waves move fast - almost 300,000 kilometers per second in the absence of

air. The round trip (up and down) is 71,572 kilometers. This distance creates a delay; just the up leg takes almost 239 milliseconds. Add in pauses to handle the traffic at the far ends (the gateways and the devices) and the ping time is 480-600 milliseconds. But many applications have no problem with this lag: live video streams, file transfers, web browsing. Even though GEO satellites are so high, voice chats are often delayed due to lags. These satellites are so far away that some network protocols' feedback loop slows with delayed responses. At the far north or south, the angle of view to the satellite is very low to the horizon, weakening the link. This increases the amount of atmosphere the signals travel through, increasing the possibility of interference. Beyond about 75 degrees, coverage is spotty or non-existent.

Near Earth, space travel occurs around 160-2,000 kilometers altitude. Below 160 km, air drag (resistance) is too strong. Above 2,000 km, space is filled with radiation from the Van Allen zones, damaging electronics. Most telecommunication networks are between 550 and 1,200 km. At 550 km, it takes 95 minutes to orbit the Earth. So, every satellite drops out from view behind the horizon in a few minutes. To provide continuous coverage, several satellites across a number of orbits are required. At least one always needs to be overhead when viewed from any point on Earth (Del Portillo et al., 2019).

Low Earth orbit is closer to Earth - 550 km - and so the signal has less distance to travel. So signals only travel for about 1.8 milliseconds - much quicker than satellites in distant geostationary orbit. Tests indicate systems such as Starlink have full round trips of less than 40 milliseconds. That's comparable to the speeds of many internet providers today. Short round trips enable people to share meetings, or play games online with minimal delays - activities where distant satellites falter. But this needs constellations of satellites in space. The cost and complexity of manufacturing, launching and coordinating the orbits of so many units adds to the cost. The satellites in space mostly last only five to seven years as they are low orbit. That's compared with a lifespan of 15 years or more for those in higher orbits. Even the rarefied atmosphere up there drags on them a little. Now and again they have to blast their thrusters. New satellites are needed to replace those lost.

A step beyond low orbits around Earth, Medium Earth Orbit (MEO) is in the 2,000-35,786-kilometer altitudes - where geostationary space begins. The U.S. GPS system zips along here

near 20,200 kilometers, as do similar satellites, such as Russia's GLONASS, Europe's Galileo or China's BeiDou. At this altitude, every satellite becomes visible for many hours over vast tracts of the Earth - all the elements needed for location services to work, but not so keen when speed is critical in delivering data, with losses of nearly 67 milliseconds per trip. Fewer attempts are made in this region for internet services, but there is one: SES's O3b network (later known as O3b mPOWER). At lower altitude (8,062 kilometers), these units reduce the lag time to 125-150 milliseconds - a significant improvement over GEO - and their coverage per unit is much larger than those closer to the earth, to link the planet, fewer crafts are needed.

The higher, the weaker. That reality affects engineers' designs of space systems. Air spreads thin the waves' energy. This is tough on distant spacecraft. That's because of the maths of waves: twice the distance, a quarter the power. High flying satellites remain feeble until they muscle up. More strength means more muscle - bigger radios, antennas. All that adds weight, cost. Down low, it's all smooth. Smaller jumps mean micro gadgets on the ground can snare signals. Tiny boxes sell quicker. Nearby circuits avoid big gear. Space's needs will be less. Cheap, lean machines, serve many users. Price is a big deal in poor countries. Up in space, organization is important to avoid signal interference. Geostationary satellites need to keep their distance - or we could get cross signals on earth. To avoid chaos, the ITU steps in, and negotiates agreements through international meetings. But in low orbit it's a different story. Swarms of tiny sats are low and sometimes bump over radio frequencies. No clear rules have emerged for dealing with these clashes - things are still evolving (Pelton, 2012).

2.2 Frequency Spectrum Management

The radio frequency spectrum is a scarce resource. While subject to international regulations, all of the connections between satellites require portions of this spectrum. The way the segments behave when transmitting signals determines which will be chosen and impacts the strength and versatility of a network. It can take considerable time to gain access to these channels, with processes not seen in other technologies.

The C band is at frequencies of about 4-8 gigahertz - most (old) geostationary comsats used that band, with 4-6 gigahertz up and 3.7-4.2 gigahertz down for fixed systems. These signals aren't much affected by rain; this is a good thing when rain hits near the equator. Weather bends less at these frequencies, so this

is a popular feature used by tropical Africa, parts of South Asia and Latin America. But wait - longer wavelengths, so dish sizes have to be larger to receive the signals. Bandwidth? Not much, due to physics. Then there's clutter from Earthly systems, particularly as some traditionally quiet airwaves are now cluttered with 5G towers in North America and bits of Europe. Local decisions have changed the freedom of satellites to use their band. Bhattacharyya and colleagues wrote about the crunch in 2020.

The start of commercial use of the Ku band in the 1980s, saw satellite equipment shrink. Smaller wavelengths mean that for a given dish, there's a much greater amplification to the signal - so equipment that required big round antennas like those found in C band networks can now use small offset dishes found in VSAT networks. These days it is common for TV signals to be broadcast directly to homes and for internet connections on boats and planes. But on overcast days with heavy rain, performance degrades markedly; rain can reduce the signal by several decibels so engineers need to ensure that links are equipped with additional power or there are intelligent changes to the data encoding so that the speed of the link is adjusted when conditions deteriorate (Pratt et al., 2019).

Plenty of room in the radio spectrum? Enter Ka band. At 17.7 to 20.2 GHz for downlinks (transmissions from satellite to Earth) and 27.5 to 30 GHz for uplinks (from Earth to satellite), this makes way for more data. Imagine half a gigahertz or more for each type of signal - that's 10 times more space than the old Ku band. The extra room helps increase the speed of data delivery of these satellites. But the weather has a greater impact. Tropical storms can obliterate more than 20 decibels of intensity. So defences have to be robust, planning astute. Innovative tweaks during data transmission now assist - a change of code type for changing conditions, swapping links between earth stations if one is rained out, processors on board making on-the-fly corrections. Such adjustments have eased nature's interference (Figure 2) so service remains stable, even in heavy downpours (Kyrgiazis et al., 2018).

Beyond Ka band, research is now on Q/V bands - 37.5-42.5 GHz for downlinks from satellite to Earth, 47.2-51.4 GHz for uplinks - for future-generation feeder links between satellites and ground stations; trials have begun in some cases. Although this very-high frequency (VHF) provides huge bandwidth, absorption by air greatly weakens the signal, so better ways to beat signal degradation need to be found. Laser inter-satellite links between satellites in space are also being considered, promising multi-terabit rates, and no congestion as with radio-based networks (Del Portillo et al., 2019).

To have signals get along in space networks, there are rules from an international body. These rules are put to use when satellite companies make agreements about plans, either one-to-one or in larger groups. Each radio-using arrangement must get registered with an international office for this. It also needs to be coordinated with others using the same frequencies, to cover the same areas without interference. This

can take years to get right; there are hard maths to do on signal spill. Negotiations may lead to agreed adjustments - they can be weaker, different frequencies, or restricted to certain areas. Large flocks of low-altitude birds with thousands of units pose new challenges compared to old-fashioned, fixed, high-altitude ones. Given their rapid motion, interference patterns change rapidly, minute by minute, hour by hour. There is little in the current regulations to help here, leaving scope for ad hoc improvements until better options are found (Bhattacharyya et al., 2020).

2.3 Modifying Signals and Strengthening Connections



Power, noise, bandwidth - they limit satellite communications. Physics' the limit, no more. It's the same for all of engineering. A framework of ideas from 1948 by Shannon stands. Designers play with how signals are shaped and encoded, getting closer and closer to the ultimate rates when noise is present. Every decision links to the real world, way above the planet. Atmospheric conditions are considered. Each loss, each gain is measured across the distances. Success is all in the hitting of the mark. Success is through skill, not chance. Success doesn't stray from nature's laws.

Phase-shift keying (PSK) and quadrature amplitude modulation (QAM) are widely used in most current satellite systems. These techniques may work with turbo codes or low-density parity-check codes (coded modulation). Together, they aid in delivering more data through narrow channels. The Digital Video Broadcasting Project (DVB) released the DVB-S2 scheme in 2005. It introduced adaptive coding and modulation for broadcasting and internet using satellites. Rather than static properties, parameters are set for each frame of transmission based on real-time measurements. When the weather is good and antennas

properly aligned, signals are up to something like high efficiency 32-APSK. This makes transmission faster with minimal errors. But when weather fuzzies up the signal or noise increases, the systems switch down - to QPSK plus basic encoding. This maintains the link, if at lower speeds. In 2014, a new version came along (DVB-S2X). It made better control of signal shaping and encoding possible. So, large data volumes could be managed more easily and accurately.

Keeping track of signal strength is a matter of keeping track of all the gains and losses from start to end. The show begins from the ground with the power put out by the station. Then there's a boost from the transmitting dish - the fancier and larger the better. Then some power is lost in space, particularly with longer distances and higher frequencies. The atmosphere (with rain) takes more energy, too. If the antennas are not pointed just right or their waves get out of phase, a further hit. The very weak message finally received by the satellite is the first step. On board, the satellite captures the faint message with its own reception equipment and boosts it. The quality of the strengthened version will depend on the amount of static on the link. Then, an identical process occurs as the signal heads back towards us. Overall link reliability is determined by combining weak links in both directions, and accounting for any in-flight repairs - made while passing through the spacecraft's electronics.

How do you know how good your satellite receiver is? Known as G/T - gain over noise temperature. The key to better signal reception is a high G/T, particularly with a small dish. To deliver signals, what counts is instead EIRP: emitted power x gain. Engineers want to make them as high as possible - but weight, power & cost constraints mean they don't. LEOs have more tradeoffs; they may have hundreds of watts, and weigh just a few kilos but still multitask data streams for thousands of hours.

3. Satellite Gets Closer and Closer Accurate Drop



3.1 High-Throughput Satellites (HTS)

A high-throughput satellite (HTS) takes the latest approach to satellite design by carving out its coverage into hundreds of smaller, overlapping and blurry beams rather than covering everything at once. From space, these mini-beams are like channels - even though they overlap a bit. Same frequencies come around and around again - but just not overlapping. It's like a cell phone tower in the city - but in space. Older birds might have broadcast widely using perhaps just a single umbrella beam, or a handful of areas, but the new models have hundreds of beams on the go. An old bird may send out a total of maybe ten to 20 billion bits per second. Imagine taking the same power and spreading it to tight bunches which quickly add up: now you're close or beyond hundreds of billions. That's all due to better geometry - not more power. This change wasn't first developed in space labs; it used rules telecoms already used. Angles change when the source is 36,000 km up in the air, but not the concept. Diagrams change, numbers don't. (Manos Kyrgiazis et al., 2018) supports the recent leap in systems.

The first big high-throughput satellite to be launched was ViaSat-1 in 2011, delivering 140 Gbps in Ka band, doubling the capacity of all other commercial satellites in service at North America at the time (Moision et al., 2016). Then came ViaSat-2, launched in 2017 to offer greater coverage, as well as even more power. Now, the trio of ViaSat-3s designed for Ka band use, to cover the Americas, Europe, the Middle East, Africa, as well as the Asia-Pacific, each with the capacity to push over 1 Tbps (Viasat, 2022). Meanwhile, EchoStar developed its Jupiter series and SES launched its O3b mPOWER - the one geostationary, the other in low orbit - to all gain similar gains in capacity.

HTS is made possible by a special antenna design. It doesn't broadcast like other satellites, but uses multiple tight beams. These focused beams are either from an array of mini-transmitters or a large dish with multiple feeds. In succession, they hit small areas of the ground. Energy is concentrated because it's not spread over a large area. Those people with small and cheap equipment below can detect the signal. It's like a magnifying glass, which brings the sun's rays to a point. This fine targeting leads to the satellites "zooming" in (Evans et al., 2005).

HTS multibeam antennas pose complex choices. As beams narrow, signals get stronger and frequencies can be re-used more; but these require complex technologies and create complex interference towards their edges. As adjacent beams overlap - each beam using different slices of the spectrum - interference is most challenging. Controlling how energy is spread out is important in keeping interference at bay: suppressing unwanted emissions and keeping different polarisations separate helps. In today's HTS setups, bandwidths are usually split into four or six parts, then sets of bands are cycled so a particular band is never used in nearby spots - a technique similar to that used by ground-based cell towers (Kyrgiazis et al., 2018).

3.2 Frequency Reuse Architectures

Key to how well an HTS system works is how often it reuses frequencies, and how often each piece of spectrum is used in the satellite's footprint. Imagine four beams where each slice of band runs in some 25 of the 100 beams at a time, spaced out so they don't overlap. With up to 1 Gbps per beam, and 25 beams using the same slices, we get 100 Gbps - no better than full isolation since each signal uses a quarter of the available range. Advantage comes via narrow beams. Beam power concentrates onto tight spots for higher local data density, and replicates across spots. Growth in efficiency results not from wider channels, but smaller spots.

In some new satellite systems, ancient fixed-frequency techniques are thrown out to make way for better signal shaping with software. Rather than fix beams at construction, they now reshuffle beam shapes in flight. Usually beam spots and shapes are fixed, once, according to dish placement - never to change again. Now with digital relay units and smart grids of antennas onboard, beam shapes may change in-flight. Beam spots can be stronger (when needed) and weaker (when not) - adjusted from earth. Such flexibility is most useful for areas of high variability. Where airplanes and ships go, where people gather - locations by fluke. So SES mPOWER and ViaSat-3 operate, delivering power as needed.

Ground-based MIMO technologies bring precoding - a technique aiding HTS to resolve signal interference. Rather than simply passing data, satellites fine-tune the amplitude and timing of waves per

beam to get signals to combine properly at addressed devices while cancelling at clashing locations. These adjustments help to boost quality of the signal across large areas, allowing closer-frequency recycling (Kyrgiazis et al., 2018). Unlike their ground-based cousins, space versions face delays; information about link quality is too slow for optimal tuning and dozens of beams increase maths needs. But work goes on apace, likely making practical use likely within a decade.

3.3 Focused Coverage using Spot Beams

One spot beam is the fundamental element of HTS zones of coverage. To its users, one spot beam is a chunk of land where a particular combination of frequency and polarization is available - typically in the form of a roundish oval, typically hundreds of kilometers across if originated from a high altitude GEO satellite, but only tens of kilometers across if originated from a low altitude LEO satellite. To satellite engineers, the same beam looks like a tight pulse of power shaped by the way the antenna radiates electromagnetic waves - either by shaping the mirrors of the antenna to focus the waves or by precisely timing and adjusting power to the elements of an antenna array.

Within a spot beam, signal strength concentrates itself over a region of the earth. This makes the square metres on the ground receive more power. So small dishes can be used to receive signals. At the centre, two things work: broadcasting strongly and a dish that's really precise. But as you move away, things aren't so clear. Away from the centre, the dish isn't as efficient. Additional noise leaks in from other beams. We can see this change around the 3 dB point - when capture drops to half its best (Pratt et al., 2019).

Spot beams divide up traffic by geography. If it's there, another can't get in the way. Rather than everyone using the same, like on older satellites, some get their own. This can be done using different frequencies or wave polarisations. This enables different parts of the globe perform differently. One section may carry huge data volumes to urban areas. Another offers minimal connectivity way out. One satellite, many uses. The beams are independent, not interfering with each other. Power passes from hand to hand, as needed. This doesn't impact that. Isolation builds flexibility. Matching needs to setups. Form follows function (Evans et al, 2005).

In the sky, how focused a beam can be on the ground is determined by the dish size. Larger dishes = smaller beams. Ka band satellites with 3-5 m diameter dishes form beams about 0.3-0.5 degrees in diameter - that's about 150 to 250 kilometers from geostationary orbit. Watch this space: new high

capacity birds. These will have larger reflectors or smart array antennas, mimicking giant antennas. Their beams? Sharper. Much sharper. Which will mean they'll reuse frequencies more aggressively, allowing more data to be transmitted where it's needed to be (Moision et al., 2016).

4. Satellite vs. Terrestrial Data Communication: Which is Best?

Satellite communication vs. terrestrial is not simply a "winner takes all" situation. Rather, consider them complementing each other, suitable for different uses, locations, as prices vary by place or use. Old-school transmission like fiber, coax cables, even cell towers have traditionally dominated urban areas where installation costs could be distributed over many users. On the other hand, satellite-based works where people can't easily erect towers (over water, hills, in the air). This balance is so familiar today, but new developments in both are changing their relationship. Though stable in the past, these days they are less so.

A huge advantage of satellite links? They reach more places than terrestrial. Not just urban areas, but the oceans as well - even the far reaches of the polar regions, unless it's too far over the poles themselves. Remote mountains, deserts, islands - locations where it doesn't make sense to build cell towers - get their signal from sky. Boats laden with cargo, cruises for fun, offshore drilling rigs for oil and gas - they all have access to the sky. Aircrafts crossing the ocean are connected only by satellites. If not, many would be lost in flight. Satellites, UN experts say, are the key to growth. For many who lack access to the normal internet - some 2.6 billion people in the world - the only solution today is machines orbiting the planet. They live in places cables and towers can't reach (and won't be laid unless they can make money). Space technologies step up, unobtrusively.

On earth, networks are cheaper, faster and have greater capacity. A single glass fiber thread carries a ton of data per second thanks to optical sleight-of-hand - more than the space web can supply to those who are connected. Add more threads or fancier equipment for almost limitless growth. Ground towers now use 5th-gen to respond in mere milliseconds but satellites are slow - even the low earth orbiting satellites. Time-sensitive activities such as gaming or stock trading are best done with ground connections, if you can get them (Bhattacharyya et al., 2020).

The difference between satellite and ground connections is in the dollars. Towers and cables cost a lot initially, but cost little to serve more people after that. Fibre optic cables buried in the ground involve digging up roads (expensive) but using cables becomes very cheap. Satellites are very expensive to get

into space in the first place: they require the construction of sophisticated equipment, and they have to be launched into orbit by rockets. Once launched, their capacity remains constant - each new user slices off a chunk of this pie. With terrestrial connections, adding a single user to the maximum number doesn't make much difference. This means that traditionally, data transmission per unit costs much more when using satellite technologies than when on land lines. The gap has been greatly reduced by new high-speed designs - but it's still there (Del Portello et al., 2019).

A single approach to reliability doesn't work for everyone. Telephone cables sometimes break if diggers damage them or storms cause floods, but again, re-routing ensures quick connections. But they can also recover fast - in under a second. But space is free of busted roads and diggers. But weather refracts radio waves (especially stronger ones) from time to time. Occasionally the sun bursts or the satellite's orbit changes. Satellites bypass earthly gear, and are great in emergencies. The military uses them too, as ground equipment can disappear at night (Pratt et al., 2019).

First of all, delays are a big deal when it comes to choosing satellites. For real-time communications - such as VoIP phone calls, video chats, fast action online games or telemedicine visits - data going back and forth is generally the most important factor. Satellites in orbits far from Earth introduce delays of about 480-600 milliseconds, putting a kink in the language when two people talk because we expect immediate replies - ideally, less than 150 milliseconds. In contrast, networks like Starlink that sit closer to the Earth have round trips of only 25 to 50 milliseconds - comparable to typical home internet services using cables on Earth. Thanks to this advantage, low-Earth satellites now do stuff their cousins in distant orbit can't do very well, particularly in places where there's no wired infrastructure (Del Portillo et al., 2019).

Let's consider security. On land, cables transmit data using fixed (controllable) systems. While hackers may be targeting the digital portion, stealing the data directly from fiber optic cables is difficult, and will be detected. Data sent to satellites is broadcast through the air, and can be "listened" to. And without good scrambling of data, good listeners with the right equipment can tune into what's being transmitted. Most user data is now secured by good locks (such as TLS or IPsec). But the channels to control the satellites? These remain insecure without vigilance (Bhattacharyya et al., 2020).

A glance at the table reveals no one winner. If fibre in the city or town warrants 5G or 6G for fast, responsive, high-data-rate applications, the sky is the limit for satellites to cover large areas, mobile

platforms or remote areas. Reliability also favours space links, too. Rather than selecting one or the other, blending them is surreptitiously growing - with data routed via the best link for the job at hand, and location. Both on earth and space companies now build systems that work with that mix.

5. Problems Sending Data From Satellites

5.1 Atmospheric Attenuation

Up in the sky, air bends radio links, rather than allowing them to pass straight through. Waves can fade, scatter, get out of alignment - all things to be considered by the engineers. Water vapour absorbs energy; clouds are a drag; raindrops are bullets. All problems affect some frequencies more than others. Regional, diurnal and seasonal variations too.

Humidity and oxygen molecules are the big absorbers of radio waves. At 60 gigahertz, oxygen absorbs so much that it's difficult to make long distances links. But below 40 gigahertz, the weakening of signals from gases is relatively predictable. Such weakening varies slightly with the elevation of the link, but is easily built into typical planning values. Rain complicates things to make issues more acute - and variable. Downpours may occur over a large area, extending over several kilometers. Fat raindrops come with a big loss of signal strength; loss depends almost entirely on the rain. Equatorial downpours might seep through paths when frequencies reach the Ka band. Rain at 25 millimeters per hour could remove five to 10 decibels from the signal strength. During more severe weather - during which rain may fall at a rate of a hundred millimeters per hour - it could be 24 decibels or more. Typically, space communication systems operate for all but a few moments of every year. To cope with relatively short but intense weather events, designers of space systems use extra margin, or change strategy in the middle of the game when fading sets in. These moves ensure streams of data don't dry up even under dark skies.

How to keep fades at bay? Spread out ground spots. Rain is often localised, so if you have two sites, 50-100km apart, chance of both being hit with big rain fades is slim. If the primary hub has problems, move traffic to the other hub - keeps the network going. Great if you want to connect satellites with big fixed sites with tons of data. But it's not much use for users on the ground. So they change how the signal is encoded - with less complex settings, they can stay connected even if there are storms. Speed goes down. Connection holds. That's the trade (ETSI, 2005).

5.2 Propagation Latency

While many people focus on propagation delay in orbital mechanics, its implications for systems design don't get as much attention. Signals travelling to geostationary orbit experience delays between 480 and 600 milliseconds one way (round trip delays are twice this) - this might seem like an arcane detail, but it's part of what makes networks work. TCP/IP is all about timing feedback, so slow responses make timing off. If each packet sent takes longer to get acknowledged, its speed slows down. Once packet loss occurs, it's harder to ramp up speed as waiting times kick in. As Bhattacharyya et al pointed out in 2020, such delays are no small thing - they change the game for protocols.

In the world of space tech, certain helpers (PEPs) have emerged to help speed up the internet with large orbit relays. These helpers capture traffic from the web at ground stations, "simulate" instant reply signals, then manage the actual delivery on sky links themselves - with smart rules to compensate for long delays. Of course, this works fine, but it disrupts regular checks and balances between devices communicating electronically. Scrambling data shuts out middlemen since hidden parts of messages can't be spoofed, conflicting with legacy setups relying on peeping into messages. Add in new setups such as walled gardens and flip-flop messaging (which mask even more of the communication) and many existing tricks go away in the night. With more scrambled formats on the rise, engineers work hard behind the scene to iron out speed tweaks without compromises to privacy guarantees of modern code layers (Del Portillo et al., 2019).

Low orbit systems reduce signal delay so much that standard internet protocols will work for all but a few things. But speed poses another timing problem - these rapid satellites are always on the move, changing the route of the connection. As a result, satellite handovers are frequent - even as often as every few minutes. Switches can potentially lead to micro-delays or bit loss if not done right. Now, new network designs anticipate changes and use forecasts to prepare for each change, holding bits temporarily and co-ordinating actions between outgoing and incoming satellites, ensuring continuity of service (Del Portillo et al., 2019).

5.3 Space junk and space safety

In the sky, high above us, it's getting crowded. Particularly up close - LEO is no longer. Teeming with small sats from companies that are putting up constellations. Already crowded with active satellites, dead boosters are on the prowl. Space has been littered with the debris of 60 years of space missions. Hit something? Consequences hit hard. Even small rocks hit hard when travelling at that speed. Things

collide - they smash to bits. Now you have a bunch of fragments, instead of just one. All the fragments move randomly. Each might hit a ship in future. This sort of snowball effect used to be theoretical. In 1978, two scientists outlined how collisions lead to collisions. And if left alone, their spiral leads to orbits that are hazardous to all who travel them.

Junk now grows up there each year. In early 2023, the U.S. had tracked some 23,000 bits larger than ten centimetres orbiting Earth, with unseen estimates putting perhaps 500,000 bits between one to ten centimetres - too small to track, but big enough to smash spacecraft - orbiting Earth. Remember that Iridium-Cosmos smash-up in 2009? That was a shower of debris. As did China's rocket smash into its own weather satellite, Fengyun 1C, six years before. Each time added thousands of lines to the catalogues (Pelton, 2012).

Attitudes towards space junk have changed. But some think this is not big enough to fit the trash. There were some suggestions from IADC on how to clean up after space missions. They suggested that equipment be removed from low orbit in 25 years. Other countries adopted these rules, and made them stricter. One change was made by the FCC in 2022. It announced US satellites need to de-orbit quicker - after five years. This attempted to keep pace with the many new satellites. Befitting the space age, SpaceX said its Starlink satellites would stick to the new time line. The satellites have plenty of power to plunge faster into the atmosphere once they're used up (Bhattacharyya et al., 2020).

At the "edge" of Earth, robots could soon catch junk. It's something that used to be on paper only. But it's starting to happen. One is from ClearSpace - a venture that started at EPFL, and is partly funded by the European Space Agency. Their plan? Launch a vehicle, ClearSpace-1, into low orbit. There, it will grab a bit of debris left over from a Vega rocket launch. The mission: to bring it down to burn up in the atmosphere. The test will take place mid-decade. But catching a tumbling target, when it's not "co-operating" is challenging. Finding it in space is tricky. Aside from the technology, funding peters out - we don't yet have a good way to pay for cleanups. So, these are still experiments. Not routine work.

5.4 Cybersecurity in Space

Recently cybersecurity threats to space-based communication systems have become a hot topic. The problem has been highlighted by high-profile attacks, particularly given the links are used for critical government services. A significant incident: as Russian attacks kicked off their full-scale invasion of Ukraine - February 24, 2022 - a cyber attack took down Viasat's KA-SAT network. This attack led to the

outage of thousands of modems in Europe. For a few days, Ukraine's defence forces, and even civilians, were cut off. Criminals exploited a vulnerability in the system and uploaded new code to the modems until they broke completely (Recorded Future, 2022).

Satellite signals are left open. Anybody with the right equipment may catch what's on its way down, without locks. Manipulating a spacecraft? It's found where commands go back and forth. Intercept the links - someone else alters the mission in flight. On the ground, there are typical cyber attacks. Hacks infiltrate like everywhere. Malware stalls work. Service is blocked by fake requests. And vulnerabilities lurk like they do in computers on the ground (Bhattacharyya et al., 2020).

Rigorous login procedures and encrypted data help secure user-data channels and control channels in satellites. Segmented networks help keep problems in check if part of a ground network fails. Close tracking of components sources minimises the problems of modified parts in factories.

Software-based ground equipment allows for more attacks - air transmissions of software updates could be intercepted, unless rigorous security measures are in place. If the remote update process isn't secure, neither is safety. In 2022, after KA-SAT, the US Cybersecurity and Infrastructure Security Agency (CISA) set guidelines on secure satellite communications. Soon after Europe's cyber team set up a framework to protect space systems.

6. Future Trends

6.1 Hooking up to 6G

Future mobile technology "6G" today is in the laboratories and conference rooms, inching towards deployment in the 2030s. Not just looking to be faster, or have lower latency than predecessors. Rather, 6G is built from the ground up to include space links. Earlier networks such as 4G or 5G only relied on satellites as back-ups. Future 6G rules designed by bodies such as 3GPP, ITU and the IMT-2030 team, however, consider sky platforms, orbiting relays and ground towers as equal. One seamless network emerges from these plans - continuous, interconnected, seamless. That's what's new.

It's hard to integrate satellite and terrestrial networks at a signal level. A 5G tower on land communicates with devices in millionths of a second - needing tight time co-ordination and rapid feedback to keep the connection up. With a low orbit satellite serving the same device comes delay - short enough to deal with, but with new time synchronisation rules. As it travels fast - around 7.5 kilometers per second - it causes the signal to shift: this needs to be corrected on board and at the user ends. The 2022 update of the global mobile standard (version 17) added space network support to 5G wireless communications, detailing

changes needed so normal 5G devices can connect via satellite. This revision makes small aircraft in low orbit into flying cell towers for coverage where towers can't reach (Bhattacharyya et al., 2020).

And phones now connect directly with Earth-circling satellites, without land antennas. Once mere speculation, it suddenly became reality with innovations such as Apple's new emergency satellite texting on the latest iPhones. Likewise, other initiatives, such as some Qualcomm chips, provide minimal text handling in emergencies. These initial offerings can handle only tiny amounts of information, not yet the speed of the internet. The next step aims to let people access the internet regularly, right through their normal phone designs. To do this, they want stronger signals to Earth, and equipment in space that's much better at listening for weak signals up. Across the globe and across nations, regulations are complex - sharing airwaves intended for cell towers from space requires discussions. Companies such as AST SpaceMobile are already in talks with current wireless companies to fairly use those waves.

Space joins in on slicing networks with 6G satellites. Not only towers - virtual lanes can even extend to orbit, directing data. A single real network can split to form multiple virtual lanes for various purposes. A rural sensor may send occasional small packets via a low-speed, low-cost satellite link. High-speed video for ships gets priority with more bandwidth. Software services in the cloud manage it all, regardless of where signals are sent or received. The European Hexa-X, Japanese post-5G, Chinese IMT-2030 and work in North America have this in mind. Combining terrestrial, aerial and space communications into a unified network management system is a big focus.

Thinking less about speed, more about power saving has begun to influence designs for future networks - even those in orbit far above Earth. Rather than just trying to get faster speeds, new standards for new technologies include power efficiency. Once launched, satellites don't consume a lot of power while operating in orbit, but launching them is exhausting: this has a big impact on the overall carbon footprint. Satellite factories are also switching up their methods: printing parts with 3D printers, sharing designs between different models, even reusing rockets as SpaceX does today - all this helps reduce resources wasted per launch. They harness sunlight with improved solar cells, and have more reliable batteries to last years without breaking down, so fewer get sent up over time.

6.2 AI-powered smart satellites



Large numbers of satellites mean lots of information to keep track of, and traditional methods of humans monitoring it are becoming overwhelmed. When dealing with a single low orbit network of thousands of satellites, a never-ending flood of data comes pouring in - information on power, temperature, location, health of equipment and signal quality. Keeping an eye on all these data points, looking for anomalies, responding quickly - when you do this for a fleet of satellites it's beyond human capability without some help from technology. Rather than just having humans spot problems, artificial intelligence (AI) systems are increasingly involved. They learn from previous experience, analysing huge volumes of data beyond human capacity. As the size of fleets expands, relying on machine intelligence goes from nice to indispensable.

The value of artificial intelligence for space missions shines when it comes to satellite maintenance. Rather than waiting for components to fail, the focus is on learning from historical data of fleets of machines in orbit to predict problems in advance using intelligent algorithms. This allows them to catch these signs before they become serious, and to make changes, such as reorienting or redirecting electricity. At SpaceX's Starlink operations centre, AI-driven systems monitor the live data from tens of thousands of satellites simultaneously. If there's a problem, it raises an alarm to human operators. Action can even be fully automatic - albeit still within limits - upon detection. With fewer unexpected events, there are fewer network outages. Satellites life expectancy improves too, with longer lifecycles than previously expected (Del Portillo et al., 2019).

In space, it's hard to keep satellites safe as more and more gets thrown into low Earth orbit. As more satellites are launched, near misses (we call them conjunctions) occur more frequently and each needs to be analysed or sometimes even to move. For Starlink, their satellites have hundreds of moves per week, mostly selected by computers. Behind the scenes, artificial intelligence scans through thousands of potential collisions, to decide which might be bad, based on risk and difficulty of changing course. Decisions are selected carefully, to line up the spacecraft to avoid danger, while not using too much energy, and not interfering with internet data down on the ground (Kessler & Cour-Palais, 1978).

Artificial intelligence has led to improvements in resource management for HTS systems. In satellites managing hundreds of targeted signals and multiple users, sharing bandwidth, satellite power and the direction of signals gets very complicated. Rather than fixed allocation tables, intelligent systems now adapt to real-time data - such as the amount of data transferred, outages due to weather, or user distribution - and make adjustments in real time. The tweaks increase the overall capacity of the network and maintain fairness of service to users. These AI systems respond swiftly to signal degradation, so they often anticipate problems. And they reveal nuanced means to allocate resources that boost the network's overall efficiency (Kyrgiazos et al., 2018).

Ultimately, artificial intelligence begins to design satellites. Not just assisting - actually suggesting options for structure and thermal control by exploring possibilities humans can't easily explore. These intelligent techniques design components that are lighter, and more able to cope with hot and cold extremes than their predecessors. But don't think "and", think: but also, these tools tweak orbits, fine-tuning constellations to specific reaches, speeds and data rates needed. While humans test hundreds in the same time, the machines test millions. Space networks get more complex, so we use more machine intelligence - insidiously intermingling with planning, operations, repairs. Over time, this drives towards self-repairing, self-organising systems, like organic webs, in the sky, like some scientists envision the sky of the future to be full of chatter.

With AI pilots for satellites in tricky situations, finger pointing is murky. Who takes responsibility for bold decisions by smart satellites? Fractions of a second dictate who or what gets control as peril looms - humans or computers? Today space experts debate such matters. The results will help playbook how smart tech operates in space.

7. Conclusion

A piece of paper by Arthur C.. started a revolution - now we now have flocks of flying data machines goes zip zip zip. Satellites no longer just sit and wait like their ancestors did, but are a mix of plodding giants, far out, and fast-moving herds, closer in. Physics keeps it together: keeping them in orbit, which radio frequencies they use so as not to collide, how signals are shaped and broadcast through the air. Ever-so-slight adjustments by engineers over the years ironed out the wrinkles, and transformed awkward start-ups into something that's cheap and effective. Thanks to these, it's often cheaper to transmit data through air, rather than wires.

What if satellites were like a camera lens? Today's High-Throughput Satellites are based on this concept. Rather than the entire country, they have lots of beams. Beams only cover a single patch of ground - even as small as a few hundred kilometers. With reuse of these frequencies in different patches, capacity soars. A decade ago such speeds were unheard-of. Sea links are still good offshore. Aircraft are seamlessly connected. And rural homes have reliable connections. Such things weren't possible until tight-beam designs came along. It's like prisms - one beam of light per section, focused. It's cost effective and efficient. To get so much data from 36,000 km away seemed impossible. Now it runs daily.

Fat air weakens signals, distractions get in the way of data, space junk orbits faster than safe trajectories can open, also dangers lurk in digital connections - all require engineers to be wary. If not controlled, junk in orbit is a ticking bomb: each new satellite increases risk even as private companies scramble to get into space. There are rules - the 5 year deadline for clean-up, advice to all on managing junk, plans to remove junk - but they may not be enough until countries co-operate, and not merely consent.

Satellites are now better linked to the terrestrial network. Third Generation Partnership Project (3GPP) NTN standard for 5G makes space tech work with cell towers. Satellites could be a natural part of 6G networks. And cell phones hooked up to space relays accelerate the trend. Once distinct worlds - space and earth - are beginning to merge. Intelligent routing makes decisions about where to route calls. Behind the scenes, artificial minds take care of what can't be done in control rooms. They learn as they go in fleets of orbiting satellites. It's all so simple when decisions are made beyond human comprehension.

Far from redundant, satellite data links are critical, even with terrestrial networks. Each year they get better, smarter, faster. Low Earth orbit systems make themselves sharper, bringing speed previously unheard of from space. High-throughput satellites approach multi-terabit speeds, and stretch the boundaries of what's possible. Software optimises control of huge swarms of space machines without breaking the bank. No longer are these systems backups waiting for cables to go down. They're beside fiber and cell towers - not behind. Their place? In the front and centre, not on the fringes.

References

Al-Hourani, A., Evans, R. J., Kandeepan, S., Moran, B., & Eltanany, H. (2016). Stochastic geometry methods for modeling uplink heterogeneous networks. *IEEE Transactions on Wireless Communications*, 15(10), 6662–6677. <https://doi.org/10.1109/TWC.2016.2586500>

Thom Stone, CSC, April 2004. Introduction to Satellite Communication Technology for NREN

Clarke, A. C. (1945). Extra-terrestrial relays: Can rocket stations give worldwide radio coverage? *Wireless World*, 51(10), 305–308.

Maral, G., & Bousquet, M. (2011). *Satellite communications systems: Systems, techniques and technology* (5th ed.). Wiley.